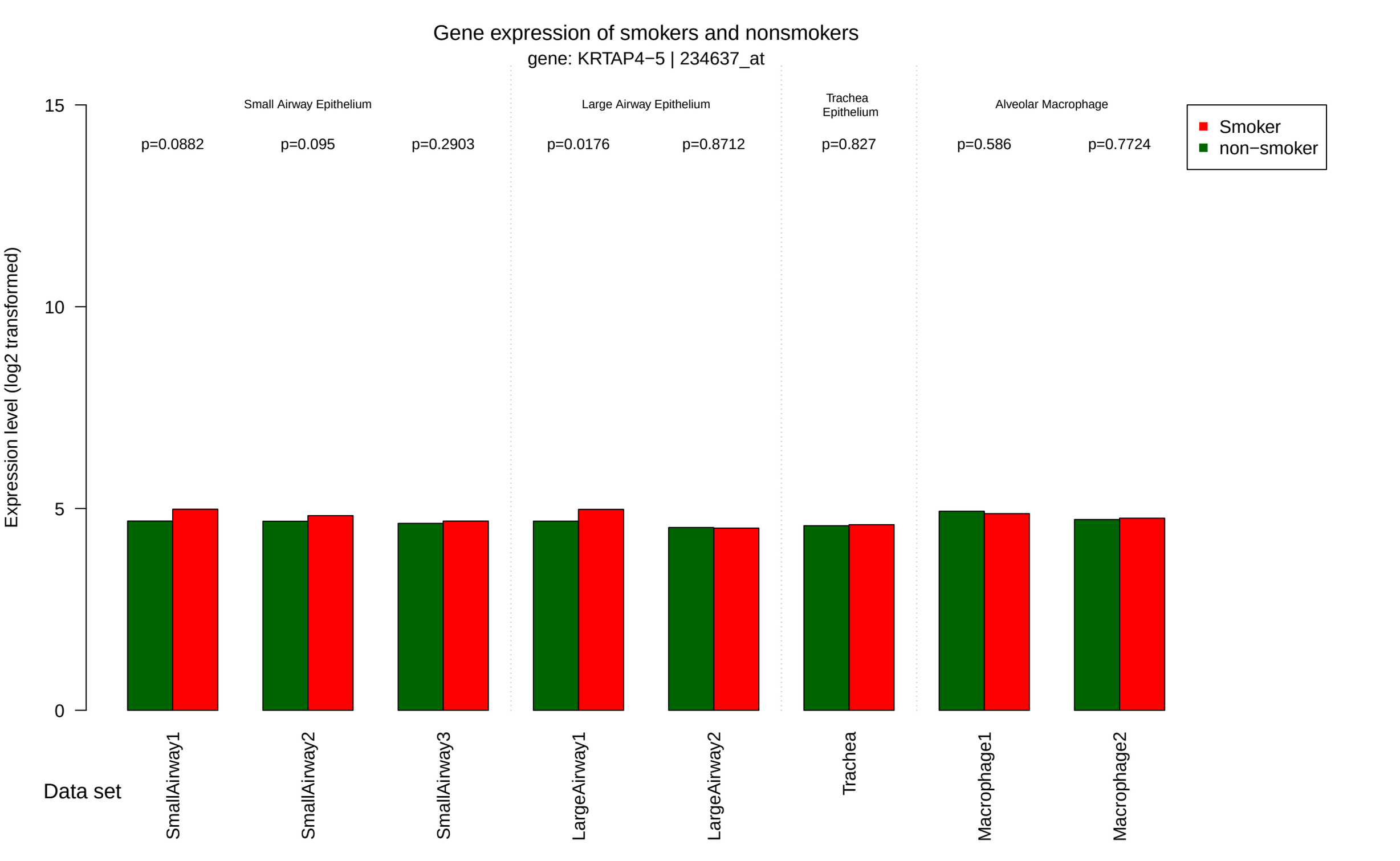
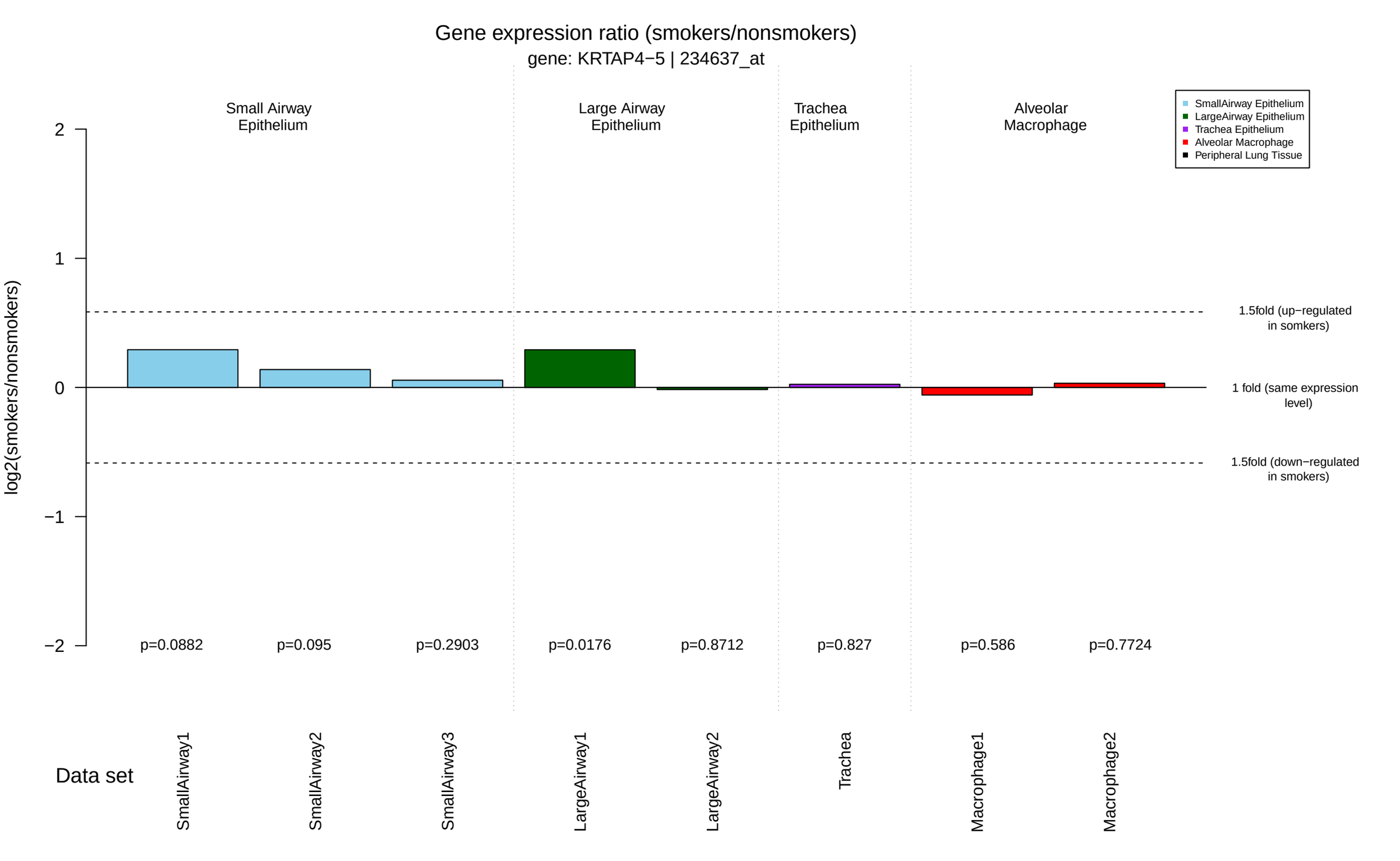
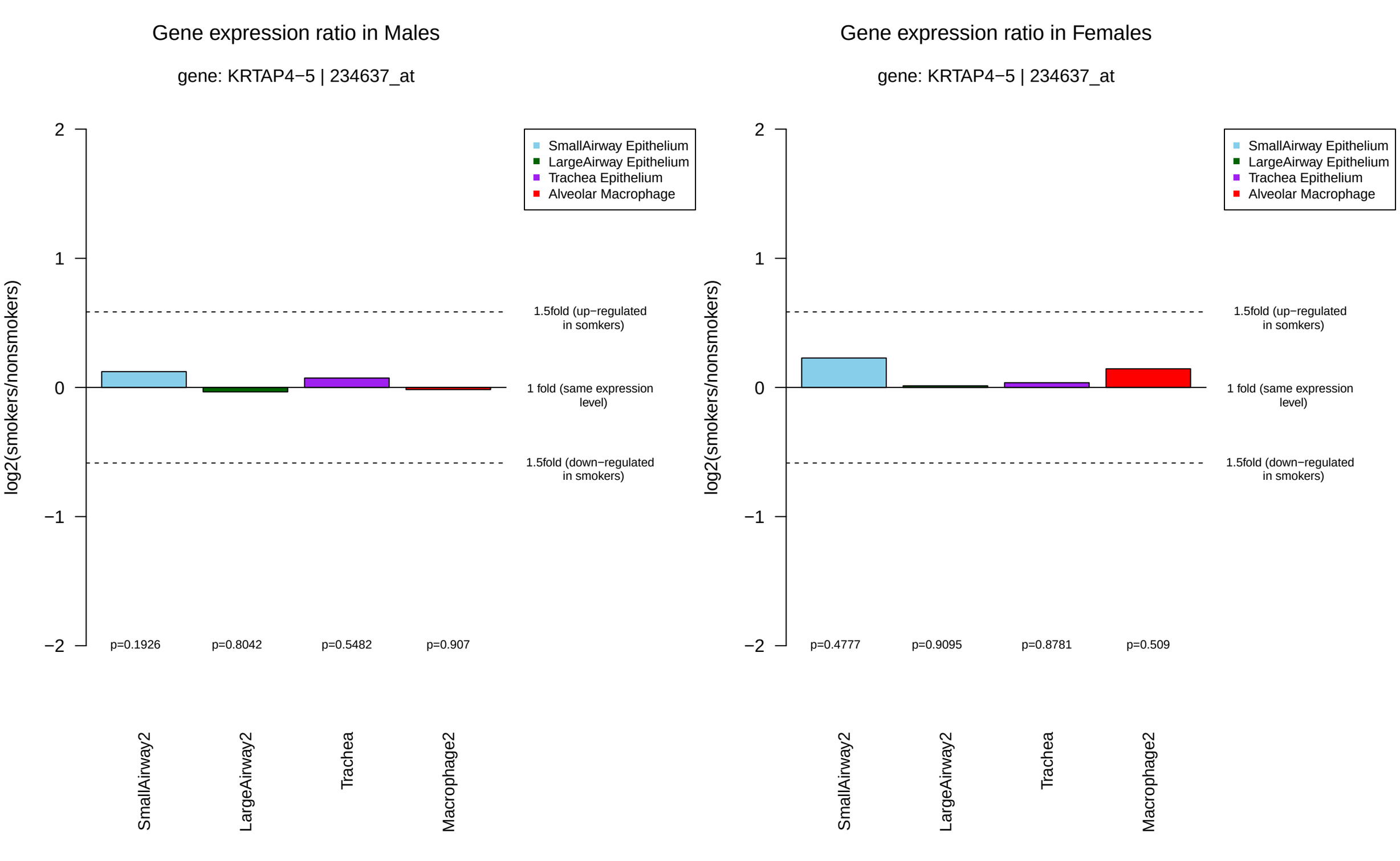
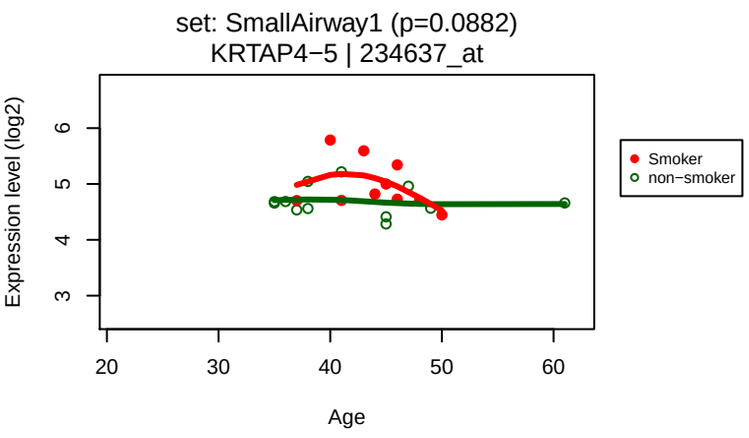
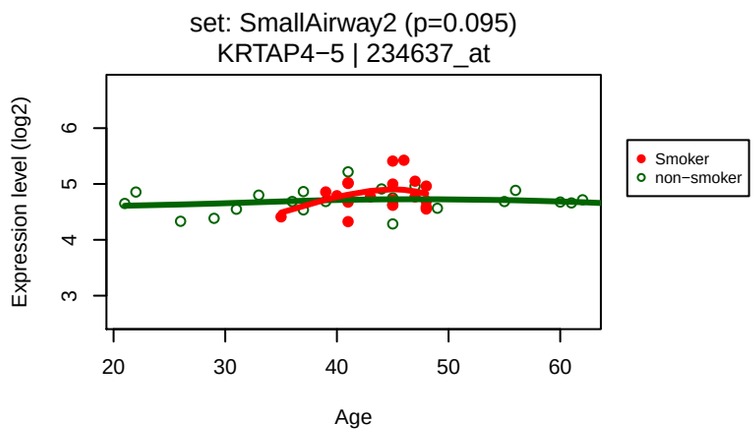
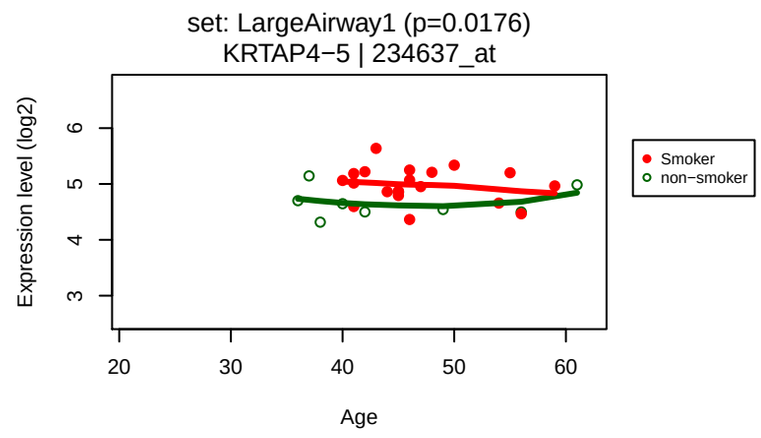

Gene expression of smokers and nonsmokers
gene: KRTAP4−5 | 234637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0882
p=0.095
p=0.2903
p=0.0176
p=0.8712
p=0.827
p=0.586
p=0.7724
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

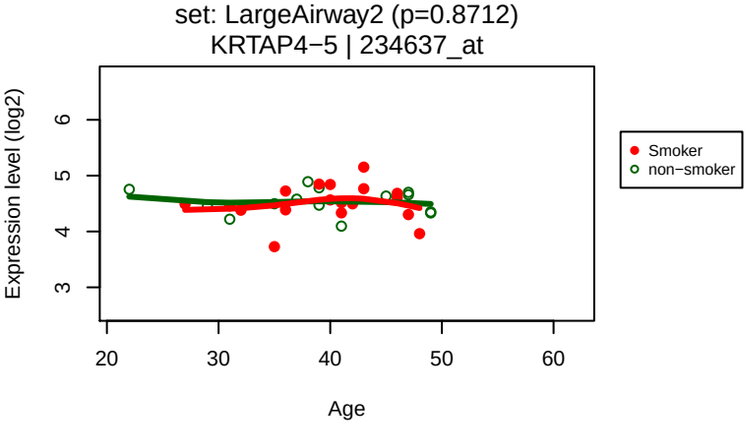

Gene expression ratio (smokers/nonsmokers)
gene: KRTAP4−5 | 234637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0882
p=0.095
p=0.2903
p=0.0176
p=0.8712
p=0.827
p=0.586
p=0.7724
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

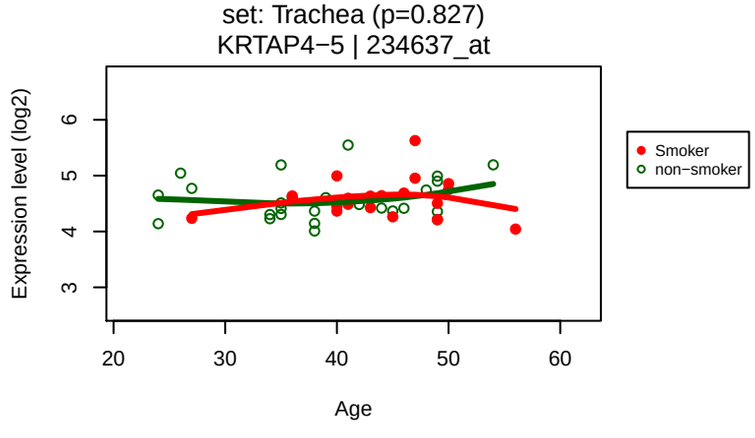

Gene expression ratio in Males
gene: KRTAP4−5 | 234637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1926
p=0.8042
p=0.5482
p=0.907
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRTAP4−5 | 234637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4777
p=0.9095
p=0.8781
p=0.509
SmallAirway2
LargeAirway2
Trachea
Macrophage2

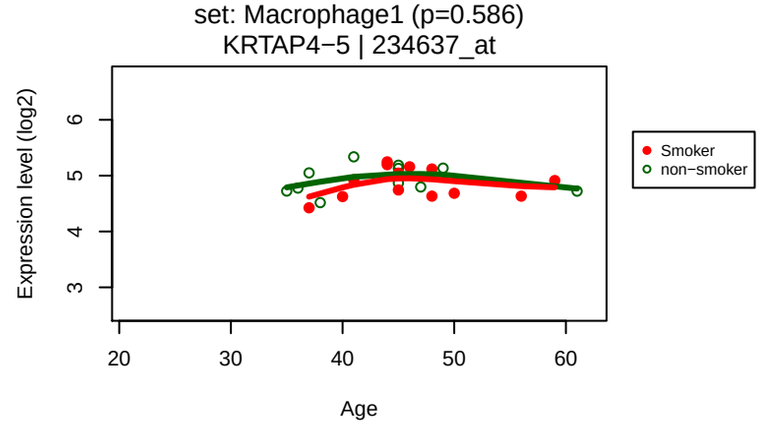

set: SmallAirway1 (p=0.0882)
KRTAP4−5 | 234637_at
Expression level (log2)
Age
Smoker
non−smoker

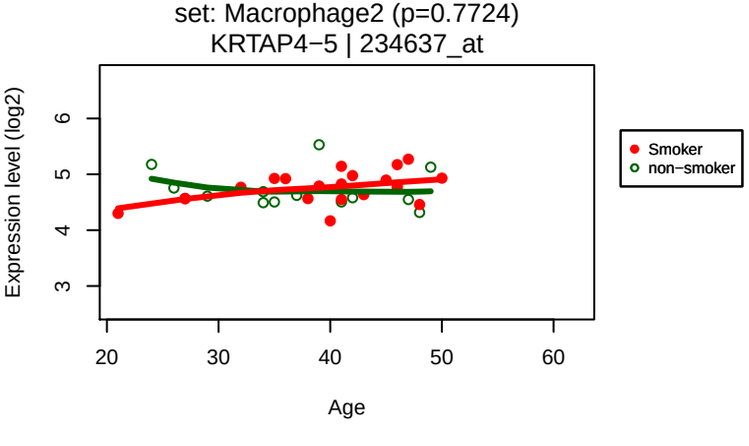

set: SmallAirway2 (p=0.095)
KRTAP4−5 | 234637_at
Expression level (log2)
Age
Smoker
non−smoker

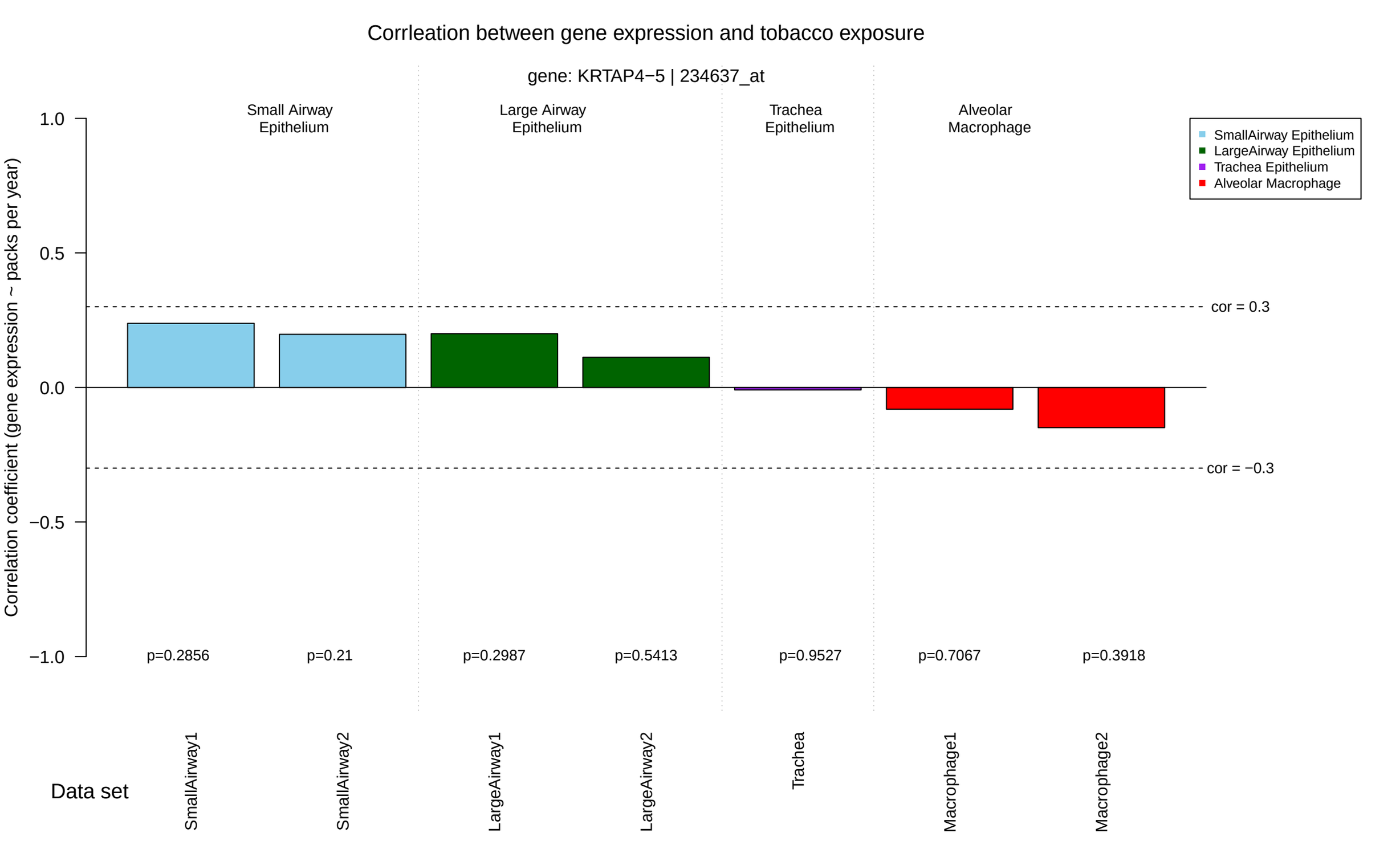

set: LargeAirway1 (p=0.0176)
KRTAP4−5 | 234637_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8712)
KRTAP4−5 | 234637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.827)
KRTAP4−5 | 234637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.586)
KRTAP4−5 | 234637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7724)
KRTAP4−5 | 234637_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRTAP4−5 | 234637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2856
p=0.21
p=0.2987
p=0.5413
p=0.9527
p=0.7067
p=0.3918
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2